Point-dependent measurement designs and the square root in the subexponential #SAT algorithm

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Abstract

A manuscript released on 26 October 2026 in the openai/math collection claims that the Exponential Time Hypothesis is false: it counts the satisfying assignments of a k-CNF on n variables in time $2^{n/(\log n)^{1/2+o(1)}}$. The square root in the exponent enters through one step of the method, a *measurement design* that separates a deformed polynomial into small independent groups. We study that step on its own. First, every design that uses a single measurement matrix at all interpolation nodes needs at least $D \cdot e^{-\Theta(R^2/Q)}$ measurements, where R is the deformation degree, Q the number of nodes and D the auxiliary dimension; so the manuscript's matrix is optimal up to a constant factor in R, and the square root is intrinsic to single-matrix designs. Second, designs whose measurements depend on the node, which the manuscript's definitions allow but its construction does not use, are equivalent inside the manuscript's swap algebra to a combinatorial problem on the cube: attach a polynomial p_v of degree $< R$ with $p_v(0) = 1$ to every vertex v of $\{0,1\}^Q$, and call the edge $\{v, v + e_t\}$ good when $p_v(x_t) + p_{v+e_t}(x_t) = 0$; the measurements needed at node t are the bad edges in direction t. Third, a *slab* family has a fraction at most $1/R$ of bad edges in every direction for every Q, which beats every single-matrix design once $Q \geq \frac{9}{8}(R+2)^2$; a family with fraction R^{-C}, $C > 4$, at $Q \approx R^{(C+2)/3}$ would improve the exponent to $n(\log n)^{-(C-1)/(C+2)+o(1)}$. Fourth, a general rank argument requires only $4^{-R}/R$, feasibility of a set of good edges has an exact dual in terms of stresses, and the natural combinatorial relaxations are exponentially weak. The core statements are verified in Lean 4 with Mathlib. All results concern the method and hold whether or not the manuscript's main theorem is correct; their novelty has not been verified.

1 Introduction

The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi asserts that 3-SAT on n variables needs time $2^{\delta n}$ for some $\delta > 0$. The openai/math manuscript *The Exponential Time Hypothesis is False* (family 378, revised 25 October 2026, [1]) claims a deterministic algorithm that counts the satisfying assignments of a k-CNF with n variables and m clauses in

$$(n+m)^{O_k(1)}\, 2^{O_k(n \log\log n/\sqrt{\log n})}$$

time and polynomial space, and $2^{O(n/\sqrt{\log n})}$ at bounded density. An upstream Lean development proves the slightly weaker exponent $n(\log\log n)^2/\sqrt{\log n}$ for a multi-tape machine [2]. An independent working group led by Impagliazzo, Paturi and Williams is examining the proof. Expositions have appeared [3, 4].

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Whatever the outcome of that review, the method raises a sharp question: *where does the square root come from, and can it be removed?* The algorithm deforms a polynomial whose value at 0 is the count, recovers the deformed polynomial from Taylor data at Q nodes, and evaluates each Taylor coefficient by separating the variables into r groups after fixing a small number of integer *measurements*. The degree R of the deformation controls the cost of the Taylor data, and the number of measurements controls the cost of the separation. In the manuscript's construction the measurement fraction decays like $e^{-R^2/(8Q)}$, so R must be of order $\sqrt{Q \log(1/\varepsilon)}$; with $Q \approx \log n / \log\log n$ forced by the auxiliary dimension, the square root follows.

Results. We isolate the measurement step and prove:

1. **A barrier for single-matrix designs** (Theorem 1, Corollary 3). Any matrix satisfying the manuscript's two admissibility conditions needs $c/D \geq \Pr[X \geq k+R] \Pr[X \leq k]$ with X binomial, hence $R \gtrsim \sqrt{Q \log(1/\varepsilon)}$. The manuscript's matrix $V_0 f(W_0)$ is optimal up to a factor of about 6 in R.

2. **A reduction for point-dependent designs** (Proposition 4). Inside the commuting swap algebra, node-dependent designs are exactly families of polynomials on the cube, and the measurements at node t are the weighted bad edges in direction t. We call the resulting optimization problem the *cube polynomial problem* $\mathrm{CP}(Q, R)$.

3. **Global formulas and the slab family** (Lemma 7, Theorem 9). There are explicit families with bad fraction at most $1/R$ in every direction, for every Q; for $Q \geq \frac{9}{8}(R+2)^2$ this beats every single-matrix design by a factor of order R.

4. **The target, and why the slab is not enough** (Proposition 11). Bad fraction R^{-C} with $C > 4$ at $Q \approx R^{(C+2)/3}$ would improve the exponent of the method.

5. **Barriers and duality** (Theorems 12, 13, 15). Every design, in or out of the swap algebra, needs only $c_t/D \geq 4^{-R}/R$; a set of good edges is feasible if and only if every stress on it has total weight zero; obstructions have at least 2^R vertices, so fractional and subcube-hitting relaxations give only exponentially small bounds; products of families preserve the bad fraction.

Small exact optima, computed by integer programming and re-verified in rational arithmetic, beat the slab in every solved case with $3 \leq R \leq Q-1$ (Section 7). The decisive asymptotic question, whether the bad fraction can be made polynomially small in R at $Q \gg R^2$, is open (Section 8).

Status. The statements concern the method, not the truth of the main theorem: they are true whether or not the manuscript's algorithm is correct. The core lemmas are machine-checked (Section 9). Our literature check (Section 10) found no prior statement of these results, but it was not a systematic review, so their novelty is **not verified**.

2 The measurement step of the algorithm

We follow the bounded-occurrence overview of [1, Section 2]; all costs below ignore polynomial factors. A formula with n variables and $M \leq 3n$ literal occurrences gives a polynomial $\#F = \sum_{x \in \{0,1\}^n} \prod_e \Phi_e(x)$. Each occurrence of X_v in clause e is replaced by a linear form $\langle \vec{P}_{e,v}(T), \vec{X} \rangle$ whose coefficients are polynomials of degree at most R in a deformation variable T, with $\vec{P}_{e,v}(0) = \delta_v$. The resulting $H(T)$ satisfies $H(0) = \#F$ and is recovered by Hermite interpolation from its Taylor

coefficients at nodes $t = 1, \dots, Q$. Each Taylor coefficient is assembled from Boolean substitutions by inclusion and exclusion. At each node the deformed clauses must be *separated*: once a small number of integer measurements of the variables are fixed, each clause should depend only on its own group of variables. The manuscript calls this an (r, τ, D)-design at the node [1, Definition 3.1]: r groups of at most $2n/r$ variables and at most $2\tau n$ measurements of ℓ_1 norm at most D. The three costs multiply to

$$\underbrace{\sum_{j \le q} 2^j \binom{n}{j}}_{\text{Taylor data, } q=\lfloor 3nR/Q \rfloor} \cdot \underbrace{(2D+1)^{2\tau n}}_{\text{measurements}} \cdot \underbrace{2^{2n/r}}_{\text{one group}}, \tag{1}$$

which is $2^{O(n[(R/Q)\log(2Q/R)+\tau\log(2D+1)+1/r])}$ [1, display (2.9)].

The auxiliary construction. The designs are built on auxiliary labels $u = (a_u, d_u) \in \mathcal{B} = [r]^Q \times [r]^Q$, so $D = |\mathcal{B}| = r^{2Q}$; for labels u, w put $E_{u,w} = \{t : a_w(t) \neq d_u(t)\}$. The construction needs a matrix $M_0 \in \mathbb{Q}^{\mathcal{B}\times\mathcal{B}}$ and a measurement matrix $C \in \mathbb{Q}^{c\times\mathcal{B}}$ with

(i) (kernel) $M_0\xi = \xi$ for every $\xi \in \ker C$, and

(ii) (support) $M_0[u, w] \neq 0 \Rightarrow |E_{u,w}| \le R - 1$,

and the counting routine requires $c \le D\tau/(3r)$. Condition (ii) keeps the degree of the replacement rows below R, and condition (i) makes the nonlocal part a function of the measurements. The manuscript uses the commuting swaps Π_s that exchange the two coordinates at position s, the complete swap $V_0 = \prod_s \Pi_s$, the operator $W_0 = \frac{1}{2}\sum_s (I - \Pi_s)$, and $M_0 = V_0 f(W_0)$, where f has degree $R - 1$ and interpolates $(-1)^j$ on a window $\mathcal{I}$ of R consecutive integers around Qp, $p = \frac{r-1}{2r}$. Then $I - M_0$ has rank $D\Pr[\mathrm{Bin}(Q, p) \notin \mathcal{I}] \le 2De^{-R^2/(8Q)}$, and its rows are measured through an orthogonal $\{0, \pm 1\}$ eigenbasis [1, Section 2.7].

3 Single-matrix designs: an isoperimetric barrier

Theorem 1 (rank barrier)**.** *Let (M_0, C) satisfy (i) and (ii), and let $A, B \subseteq [r]^Q$ with $\mathrm{dist}_H(a, b) \ge R$ for all $a \in A$, $b \in B$. Then $c \ge \mathrm{rank}(I - M_0) \ge |A|\,|B|$.*

Proof. By (i), $\ker C \subseteq \ker(I - M_0)$, so $\mathrm{rank}(I - M_0) \le \mathrm{rank}\, C \le c$. Let $U = \{u : a_u \in A,\ d_u \in B\}$. For $u, w \in U$, $|E_{u,w}| = \mathrm{dist}_H(d_u, a_w) \ge R$, so $M_0[u, w] = 0$ by (ii). The principal submatrix of $I - M_0$ on U is the identity, so $\mathrm{rank}(I - M_0) \ge |U| = |A||B|$. □

Corollary 2 (binomial form)**.** *Let $h = \lfloor r/2 \rfloor$ and $X \sim \mathrm{Bin}(Q, h/r)$. For every integer k, $c/D \ge \Pr[X \ge k + R] \cdot \Pr[X \le k]$.*

Proof. Fix $H \subseteq [r]$ with $|H| = h$ and let $N_H(a)$ count the positions with $a(t) \in H$. Then $\mathrm{dist}_H(a, b) \ge N_H(a) - N_H(b)$. Apply Theorem 1 to $A = \{N_H \ge k + R\}$ and $B = \{N_H \le k\}$ and divide by $D = r^{2Q}$. □

Corollary 3 (necessary degree)**.** *If $c \le \varepsilon D$ then $(R + 1)^2 \ge Q\, p_h(1 - p_h) \ln \frac{1}{2(Q+1)\varepsilon}$ with $p_h = \lfloor r/2 \rfloor / r \in [\frac{1}{3}, \frac{1}{2}]$ (provided $\lceil Qp_h \rceil + R \le Q$; otherwise $R > Q/2 - 1$).*

Proof. Take $k = \lceil Qp_h \rceil$, so $\Pr[X \le k] \ge \frac{1}{2}$ since a binomial median lies in $\{\lfloor Qp_h \rfloor, \lceil Qp_h \rceil\}$. For $m = k + R$, $\Pr[X \ge m] \ge \binom{Q}{m} p_h^m (1-p_h)^{Q-m} \ge \frac{1}{Q+1} e^{-Q\,\mathrm{KL}(m/Q \,\|\, p_h)}$, and $\mathrm{KL}(q\|p) \le (q-p)^2/(p(1-p))$ with $m - Qp_h \le R + 1$. □

The manuscript achieves $c \le D\tau/(3r)$ with $R \approx \sqrt{8Q \ln(6r/\tau)}$, while Corollary 3 forces $R \gtrsim \sqrt{\frac{2}{9} Q \ln \frac{3r}{2(Q+1)\tau}}$. So in the manuscript's regime ($\ln(r/\tau) \gg \ln Q$), **no matrix satisfying (i) and (ii), in or out of the swap algebra, improves on $V_0 f(W_0)$ by more than a factor of about 6 in R**, and the square root is forced for single-matrix designs. For the swap algebra and the uniform measure, the rank identity of O'Donnell [3, Lemma 4.1] combined with the sign-error bound of Aspnes, Beigel, Furst and Rudich [5] gives a similar conclusion; Theorem 1 needs neither.

4 Point-dependent designs and the cube polynomial problem

Definition 3.1 of [1] is a design *at a node*: nothing prevents the measurement matrices from depending on t. Theorem 1 does not apply to such designs, because the nonlocal part may differ from node to node. We now describe them inside the swap algebra.

Each position s splits $\mathbb{Q}^{[r]\times[r]}$ into the swap-symmetric part (dimension $r(r+1)/2$) and the antisymmetric part (dimension $r(r-1)/2$). A product eigenvector has a type $\varepsilon \in \{0,1\}^Q$ ($\varepsilon_s = 1$ when antisymmetric at s), and $\Pi_S = \prod_{s\in S} \Pi_s$ acts on it by $\chi_S(\varepsilon) = (-1)^{|S\cap\varepsilon|}$. Under the uniform measure on labels the type has the product law μ_p, $p = \frac{r-1}{2r}$.

A *swap design of degree* $< R$ is $P(T) = \sum_{S\subseteq[Q]} c_S(T)\Pi_S$ with $P(0) = I$ and $\deg c_S < R$. At node t the row of label u may read freely the labels w with $a_w(t) = d_u(t)$; the rest of $P(t)$ is the nonlocal part N_t, and $\operatorname{rank} N_t$ measurements are necessary and, through the eigenbasis, sufficient.

Proposition 4 (reduction)**.** *Put $p_\varepsilon(T) = \sum_S c_S(T)\chi_S(\varepsilon)$ for $\varepsilon \in \{0,1\}^Q$. Then $P(0) = I$ if and only if $p_\varepsilon(0) = 1$ for all ε; $\deg P < R$ if and only if $\deg p_\varepsilon < R$ for all ε; and for every node t*

$$\frac{\operatorname{rank} N_t}{D} = \frac{r-1}{r}\,\mu_p\big(\{\varepsilon' :\ p_{\varepsilon'}(x_t) + p_{\varepsilon'+e_t}(x_t) \ne 0\}\big),$$

where ε' ranges over types with $\varepsilon'_t = 0$ and μ_p is the product measure on the other $Q-1$ coordinates.

Proof. The first two statements are Fourier inversion on $\{0,1\}^Q$. The entry (u,w) of Π_S is nonzero only for $w = \mathrm{swap}_S(u)$. If $t \in S$ then $a_w(t) = d_u(t)$ and the entry is local. If $t \notin S$ then $a_w(t) = a_u(t)$, which equals $d_u(t)$ exactly when u is diagonal at t. Hence $N_t = D_t \sum_{S\not\ni t} c_S(x_t)\Pi_S$ with D_t the projector onto labels off-diagonal at t. The projector commutes with every swap, has rank $r(r-1)$ at position t, and the remaining factor is diagonal in the product eigenbasis with eigenvalue $\sum_{S\not\ni t} c_S(x_t)\chi_S(\varepsilon') = \frac{1}{2}(p_{\varepsilon'}(x_t) + p_{\varepsilon'+e_t}(x_t))$ on the eigenvectors of type ε' at the other positions. Counting these eigenvectors gives the formula. □

We checked the rank formula exactly against the actual matrices $P(T)$, computed modulo two 31-bit primes, for $(r,Q,R) \in \{(2,3,2),(2,3,3),(3,2,2),(2,4,2),(2,4,3),(3,3,2)\}$ and several families each.

Definition 5 (cube polynomial problem)**.** Fix distinct nonzero nodes $x_1,\dots,x_Q$. A *family* assigns to every vertex $v \in \{0,1\}^Q$ a polynomial p_v of degree $< R$ with $p_v(0) = 1$. The edge $\{v, v+e_t\}$ is *bad* if $p_v(x_t) + p_{v+e_t}(x_t) \ne 0$. The value of a family is $\beta = \max_t \beta_t$, where β_t is the measure of the bad edges in direction t (the measure of their endpoints with $v_t = 0$). $\mathrm{CP}(Q,R)$ asks for the least value $\beta^*(Q,R)$.

By Proposition 4, a family of value β for the measure μ_p is a point-dependent swap design with $c_t/D \le \frac{r-1}{r}\beta$ at every node. The manuscript's matrix is the *window family* $p_\varepsilon = 1 - \sigma_\varepsilon f(|\varepsilon|) + \sigma_\varepsilon G(T,\varepsilon)$, $\sigma_\varepsilon = (-1)^{|\varepsilon|}$, with G the global formula of Lemma 7 below: its good edges are those with both weights in the window, and its value is the binomial tail outside the window.

Proposition 6 (subcube obstruction)**.** *For any R distinct directions, every R-dimensional subcube in those directions contains a bad edge. Consequently $\sum_{t\in T}\beta_t \ge \min(p,1-p)^{R-1}$ for every set T of R directions, and $\beta^*(Q,R) \ge 2^{1-R}/R$ for the uniform measure.*

Proof. Let $\ell_i(0)$ be the Lagrange weights of the R nodes. For every vertex v of the subcube, $1 = p_v(0) = \sum_i \ell_i(0) p_v(x_{t_i})$. Sum over the 2^R vertices: the left side is 2^R. For each i, the inner sum $\sum_v p_v(x_{t_i})$ splits into the edges of direction t_i and vanishes if they are all good. The second statement follows by summing over the bases of the subcubes. □

5 Global formulas and the slab family

Lemma 7 (global formulas)**.** *For coefficients γ_S ($S \subseteq [Q]$) and distinct nonzero nodes put*

$$G(T,\varepsilon) = \sum_S \gamma_S\, \chi_S(\varepsilon) \prod_{s\in S}\Big(1 - \frac{T}{x_s}\Big).$$

Then $G(x_t,\varepsilon) = G(x_t,\varepsilon+e_t)$ for every vertex and direction, $\deg_T G \le \max\{|S| : \gamma_S \ne 0\}$, and $G(0,\varepsilon) = \sum_S \gamma_S \chi_S(\varepsilon)$.

Proof. Terms with $t \in S$ vanish at $T = x_t$, and terms with $t \notin S$ satisfy $\chi_S(\varepsilon + e_t) = \chi_S(\varepsilon)$. □

So $p_\varepsilon = \sigma_\varepsilon G(\cdot,\varepsilon)$ makes every edge good, and the only defect is the normalization $p_\varepsilon(0) = \sigma_\varepsilon g(\varepsilon)$ with $g = \sum_S \gamma_S \chi_S$ of Fourier degree $< R$. Conversely, if every edge of the whole cube is good then $\sigma_\varepsilon p_\varepsilon$ is a global formula: the Fourier coefficient $\widehat{G}_S(T)$ vanishes at x_t for every $t \in S$. Since parity has Fourier degree Q, no all-good family exists for $Q \ge R$.

Lemma 8 (symmetric functions have low degree)**.** *Over a field of characteristic 0, if $\deg f \le d$ then $\varepsilon \mapsto f(|\varepsilon|)$ is a combination of the characters χ_S with $|S| \le d$.*

Proof. $|\varepsilon| = \sum_s \frac{1-\chi_{\{s\}}(\varepsilon)}{2}$, and multiplication by $\chi_{\{s\}}$ maps χ_S to $\chi_{S\triangle\{s\}}$, raising the degree by at most one. □

Theorem 9 (slab family)**.** *Fix $R \ge 1$ and an offset s_0. Cut the weights $0,\dots,Q$ into slabs $\{w : \lfloor (w+s_0)/R \rfloor = k\}$ of at most R consecutive values. On slab k let f_k interpolate $(-1)^w$ (degree $< R$), let $\gamma^{(k)}$ be the Fourier coefficients of $f_k(|\cdot|)$ given by Lemma 8, and put $p_\varepsilon = \sigma_\varepsilon G_{\gamma^{(k)}}(\cdot,\varepsilon)$ for $|\varepsilon|$ in slab k. Then $\deg p_\varepsilon < R$, $p_\varepsilon(0) = 1$, every edge inside a slab is good, and for some $s_0 < R$, for every direction t and every $Q \ge 1$, at most $2^{Q-1}/R$ edges in direction t are bad. Averaging over s_0 gives $\mu_p(\text{bad in direction } t) \le 1/R$ for every product measure.*

Proof. Degree and normalization follow from Lemmas 7 and 8. An edge $\{\varepsilon, \varepsilon+e_t\}$ with $\varepsilon_t = 0$ joins weights w and $w+1$; it is good when both lie in one slab. The edges in direction t between weights w and $w+1$ number $\binom{Q-1}{w}$, and each w is the top of a slab for exactly one offset modulo R. Summing over the R offsets counts every edge in direction t at most once, so some offset has at most $2^{Q-1}/R$ bad edges in direction t; the same averaging works simultaneously for all directions since the count does not depend on t. □

Corollary 10 (point dependence helps)**.** *If $Q \ge \frac{9}{8}(R+2)^2$ and Q exceeds an absolute constant, every single-matrix design has $c/D \ge 0.01$, while the slab design has $c_t/D \le \frac{r-1}{r} \cdot \frac{1}{R}$ at every node.*

Proof. In Corollary 2 take $k = \lceil Qp_h \rceil$, so $\Pr[X \le k] \ge \frac{1}{2}$. With $\sigma = \sqrt{Qp_h(1-p_h)} \ge \sqrt{2Q/9}$, the hypothesis gives $k + R \le Qp_h + 2\sigma$, and the Berry and Esseen theorem gives $\Pr[X \ge Qp_h + 2\sigma] \ge \Phi(-2) - O(1/\sigma) \ge 0.02$ for large Q. □

Proposition 11 (improvement criterion)**.** *Assume the manuscript's cost analysis (1), its constraints $12D^2 \le n$ and $2Qr^2 \le n$, and its requirement $c_t/D \le \tau/(3r)$ at every node, applied with point-dependent designs. If for some $C > 4$ there are families with $\beta \le R^{-C}$ at $Q \approx R^{(C+2)/3}$ for all large R, then the exponent of the method improves to $n(\log n)^{-(C-1)/(C+2)+o(1)}$, which is $o(n/(\log n)^{1/2})$.*

Derivation. Write $\lambda = \log_2 n$ and take $Q \approx \lambda/(4 \log_2 r)$, the largest value allowed by $12D^2 \le n$. With $\tau = 3(r-1)\beta$ the exponent per variable is about

$$\tfrac{3R}{Q} \log_2 \tfrac{Q}{R} + 12(r-1)\beta Q \log_2 r + \tfrac{2}{r} \;\approx\; \tfrac{R}{Q} \log \tfrac{Q}{R} + 3r\lambda\beta + \tfrac{2}{r}.$$

Put $r = \lambda^a$ and $R = \lambda^b$. The first and last terms balance at $b = 1 - a$, and the middle term is at most the last when $bC \ge 2a + 1$, that is $a \le (C-1)/(C+2)$. The exponent is then $\lambda^{-(C-1)/(C+2)}$ up to logarithmic factors, at $R = \lambda^{3/(C+2)}$ and $Q \approx \lambda \approx R^{(C+2)/3}$. It beats $\lambda^{-1/2}$ exactly when $C > 4$. □

The slab has $C = 1$, which gives nothing; the window family is excellent at fixed ratio Q/R but needs $R \gtrsim \sqrt{Q \log Q}$. A family that beats both by a polynomial factor in the regime $Q \approx R^{2+}$ is what an improvement needs. If β could be made $2^{-\Theta(R)}$, the same computation with $R = \Theta(\log \log n)$ would give $n(\log \log n)^{O(1)}/\log n$.

6 Barriers and duality

Theorem 12 (general rank barrier)**.** *Let $P(T)$ be any $\mathcal{B} \times \mathcal{B}$ matrix of polynomials with $\deg P < R$ and $P(0) = I$, not necessarily in the swap algebra. Let $x_1, \dots, x_R$ be distinct nodes, node i using the groups of position $\pi(i)$, with π injective, and split $P(x_i) = L_i + N_i$ where $L_i[u, w] \ne 0$ implies $a_w(\pi(i)) = d_u(\pi(i))$. For product sets $A = \prod_t S_t^A$ and $B = \prod_t S_t^B$ with $S^A_{\pi(i)} \cap S^B_{\pi(i)} = \varnothing$ for all i,*

$$|A|\,|B| \le \sum_{i=1}^{R} \operatorname{rank} N_i, \qquad \textit{hence} \qquad \max_i \frac{\operatorname{rank} N_i}{D} \ge \frac{1}{R}\Big(\frac{h(r-h)}{r^2}\Big)^R,$$

which is $4^{-R}/R$ for even r and $h = r/2$.

Proof. Lagrange extrapolation gives $I = P(0) = \sum_i \lambda_i P(x_i)$, hence $\sum_i \lambda_i N_i = I - \sum_i \lambda_i L_i$. On $U = \{u : a_u \in A,\ d_u \in B\}$ every L_i vanishes, so the principal submatrix of $\sum_i \lambda_i N_i$ on U is the identity and $|U| \le \sum_i \operatorname{rank} N_i$. □

Plugging $c_t/D \approx 4^{-R}$ into the derivation of Proposition 11 shows that no argument of this kind can force more than $R = \Theta(\log \log n)$: the square root is a property of the single-matrix construction, not of the measurement framework.

Theorem 13 (stress duality)**.** *Put $\xi_t = (x_t, x_t^2, \dots, x_t^{R-1})$. For a set G of edges, a* stress *is a weighting $w : G \to \mathbb{Q}$ with $\sum_{t:\, e(v,t)\in G} w_{e(v,t)}\xi_t = 0$ at every vertex v. There is a family in which every edge of G is good if and only if every stress on G has total weight $\sum_e w_e = 0$.*

Proof. If a family exists, vanishing moments give $\sum_t w_{e(v,t)} p_v(x_t) = (\sum_t w_{e(v,t)}) p_v(0)$ at each vertex. Summing over vertices, the right side is $2\sum_e w_e$ and the left side is $\sum_e w_e(p_u(x_t) + p_{u+e_t}(x_t)) = 0$. Conversely, the conditions form a linear system in the coefficients of the p_v; by linear duality it is unsolvable exactly when some combination has zero left side and nonzero right side, and the multipliers of such a combination are a stress with nonzero total. □

Corollary 14 (obstructions are large; relaxations are weak)**.** *Every stress with nonzero total is supported on at least 2^R vertices. Consequently the fractional relaxation of $\mathrm{CP}(Q, R)$ over obstructions has value at most $2^{1-R}/R$, and the subcube-hitting relaxation has value at most $2^{-R/2+o(R)}$ by the constructions of Alon [8] and of Ellis, Ivan and Leader [7]. A polynomial lower bound $\beta^* \geq R^{-O(1)}$ must therefore use the algebra of the moment curve, not only the combinatorics of subcubes.*

Proof. Any $R-1$ of the vectors ξ_t are independent (Vandermonde), so every vertex touched by a nonzero stress carries at least R weighted edges. A subgraph of the cube of minimum degree at least R has at least 2^R vertices (split along a coordinate and induct). The fractional bound follows by giving weight $1/(R2^{R-1})$ to every edge, and the hitting bound by taking all edges at a vertex set of density $2^{-R/2+o(R)}$ that meets every R-subcube. □

Theorem 15 (products)**.** *If (p_v) is a family on Q_1 directions of degree $< R_1$ and (s_u) a family on Q_2 other directions of degree $< R_2$, then $p_{(v,u)} = p_v s_u$ is a family on $Q_1 + Q_2$ directions of degree $< R_1 + R_2 - 1$ in which an edge is good whenever its projection is good. Hence $\beta^*(Q_1 + Q_2, R_1 + R_2 - 1) \leq \max(\beta^*(Q_1, R_1), \beta^*(Q_2, R_2))$.*

Proof. If $p(y) + p'(y) = 0$ then $(ps)(y) + (p's)(y) = 0$. □

Products preserve the ratio Q/R and the bad fraction, so they cannot reach the regime $Q \gg R^2$ that Proposition 11 needs; they do turn small optima into infinite families, for instance $\beta \leq 1/8$ at $(Q, R) = (4m, 2m+1)$ from the optimum at $(4, 3)$.

7 Small exact cases

We computed $\mathrm{CP}(Q, R)$ for the nodes $x_t = t/Q$ and the uniform measure with a mixed-integer program (HiGHS through SciPy, 240 seconds per instance), re-checking every reported good-edge set in exact rational arithmetic. Table 1 gives the number of bad edges, in total and in the worst direction (out of 2^{Q-1}). Entries with "$\leq$" hit the time limit; the bracket is the solver's proven lower bound.

Three observations, all empirical: realizability is a real constraint (at $(5, 2)$ the proven lower bound exceeds the hitting optimum); for $R = Q - 1$ the optimum is four bad edges in total and one per direction; and the slab is beaten in every solved case with $3 \leq R \leq Q - 1$, and at $(5, 2)$, where a family with exactly 7 bad edges in every direction exists. The optimal families are not unions of global formulas on level sets. Whether their advantage over the slab grows with R at large ratio Q/R is the question below.

Q	R	subcube hitting total / max	CP optimum total / max	slab total / max	window total / max
3	2	3 / 1	4 / 2	6 / 2	6 / 2
4	2	8 / 2	12 / 4	16 / 4	20 / 5
4	3	3 / 1	4 / 1	8 / 2	8 / 2
5	2	24 / 5	≤ 32 [25] / ≤ 7 [5]	40 / 8	50 / 10
5	3	8 / 2	≤ 14 [8] / ≤ 4 [2]	25 / 5	30 / 6
5	4	3 / 1	4 / 1	10 / 2	10 / 2
6	3	≤ 22 / 4	≤ 60 [9] / ≤ 12 [2]	60 / 10	72 / 12
6	4	8 / 2	≤ 18 [4] / ≤ 5 [1]	36 / 6	42 / 7
6	5	3 / 1	4 / 1	12 / 2	12 / 2

Table 1: Bad edges in small cases of $\mathrm{CP}(Q, R)$ (nodes t/Q, uniform measure).

8 Open problems

Open problem 1 (the decisive question)**.** Determine the growth of $\beta^*(Q, R)$ for $Q \geq R^{2+\delta}$. A family with $\beta \leq R^{-C}$, $C > 4$, at $Q \approx R^{(C+2)/3}$ improves the exponent of the method (Proposition 11); a proof that $\beta^*(Q, R) \geq R^{-4-o(1)}$ there shows that no point-dependent swap design beats the square root.

Open problem 2 (the infinite-ratio limit)**.** The direction-averaged optimum is non-decreasing in Q (restrict to the two faces of a new coordinate). Is $\lim_{Q\to\infty} \beta^*(Q, R)$ of order $1/R$, so that the slab is asymptotically optimal, or exponentially small in R?

Open problem 3 (region-based designs)**.** Suppose the good edges are exactly the edges inside the parts of a partition of the cube, each part carrying one global formula. Each part is then a set on which parity agrees with a polynomial of degree $< R$, of measure at most about $R\sqrt{2/(\pi Q)}$ by the sign-error bound [5]. Prove that such designs have $\beta \geq c/R$, an isoperimetric statement about parity-interpolable sets.

Open problem 4 (linear polynomials)**.** For $R = 2$ the good edges form a C_4-free subgraph of the cube, so asymptotically at least $1 - 0.6068$ of the edges are bad [9]; for generic nodes the condition is that every cycle of good edges is balanced for the voltages $-2/x_t$. Is the optimal bad fraction $\frac{1}{2} - o(1)$ for generic nodes, and do special nodes (such as $t/5$ at $Q = 5$) help asymptotically?

Open problem 5 (realizing Turán constructions)**.** The vertex sets of density $2^{-R/2+o(R)}$ meeting every R-subcube [8, 7] are built layer by layer. Is the set of edges at such a vertex set the bad set of some family? A positive answer for a single moderate R would beat the slab there.

9 Formal verification

The following statements are proved in the Lake project `EthConsequences` (Lean 4.34.1, Mathlib at a pinned revision). The build passes with `lake --wfail build` and the only axioms used are `Classical.choice`, `Quot.sound` and `propext`; the certificate records file hashes and all 45 checked declarations.

Result	Lean declarations
Theorem 1, Corollary 2	`RankBarrier.barrier`, `barrier_threshold`, `count_le_count_add_hammingDist`
Proposition 6 (the full R-cube)	`Subcube.no_good_subcube`, `eval_zero_eq_sum`
Lemmas 7, 8, Theorem 9	`SlabDesign.G_edge`, `natDegree_G_le`, `G_eval_zero`, `exists_gamma`, `slab_edge_good`, `slab_eval_zero`, `slab_natDegree`, `slab_design_bound`
Theorem 12	`PointBarrier.point_dependent_barrier`, `label_barrier`, `label_barrier_explicit`
Theorem 13, soundness direction	`Stress.stress_total_eq_zero`, `moment_collapse`, `sum_edgeW`
Theorem 15	`Stress.product_edge_good`, `natDegree_product_le`, `product_eval_zero`

Not formalized: the reduction (Proposition 4, checked on actual matrices), the binomial estimates in Corollaries 3 and 10, the completeness direction of Theorem 13, Corollary 14, the derivation in Proposition 11, and the manuscript's own analysis, on which Proposition 11 is conditional.

10 Status, scope and novelty

- **Scope.** The barriers cover designs built, like the manuscript's Section 4, on the label space $([r] \times [r])^Q$ with the group rule $a_w(t) = d_u(t)$. They say nothing about algorithms of other shapes, nor about the manuscript's Appendix C, whose square root comes from a partition count of 2-adic modes.
- **Dependence on the main theorem.** None of the theorems depends on it. Proposition 11 is conditional on the manuscript's cost analysis.
- **Inspection.** We read the manuscript's Sections 1 and 2, the statements of Section 3, Sections 4.1 to 4.3 and Appendices C and D in full from the openai/math repository at commit `7ef5c0cc5`; this is not a proof audit, and the upstream Lean development was not rebuilt.
- **Novelty.** A web-based literature check on 29 October 2026 found the expositions [3, 4], which discuss no barrier and no point-dependent designs, and the closest combinatorics in hypercube Turán problems [6, 7, 8, 9] and in the interpolation and probabilistic degree of parity [5, 10, 11]. The check was not systematic, so novelty is **not verified**. A reported independent refutation of ETH by another group, known to us only from a press report, has not been inspected.

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